



Promotion Case Study

Assessment for Senior Staff Engineer

eClaims

Electronic Claims Processing^{v1.0}

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1 Introduction

“YCompany” is one of the renowned name in Insurance providers in industry providing auto insurance to customers in US. The company's product offerings include auto, term and life insurance.

This document describes a problem that “YCompany” is currently facing. The company is planning to modernize claims processing part of the application.

This case study is technology agnostic. It is expected that you to go through the problem and based on your understanding propose solutions to various questions at the end of this document.

2 Problem Statement

“YCompany” servers more than 200 million customers across various geographies. YCompany being a key player in its business area has been facing challenges with the manual process currently in place. YCompany sells insurance policy, the process is electronic and well organized. The claim processing part of the business is still manual and heavily driven of manual touch points, which causes a lot of pain for the customer. Being a manual process for claims processing the customer claim settlement time is large and claims are sent out as cheque. This has resulted in customer dissatisfaction and YCompany losing customer confidence.

There is an additional overhead involved with claims adjuster process being manual, currently claims adjuster must submit manual assessment from field to back office, the back office then validated the assessment against the claims submitted, which is both cost inefficient and time consuming. This being a manual process customer has no way of tracking the state of claim and being well informed on when and how much claim payment they can accept. As the claims data is not collected electronically YCompany has no way of running analytics on the data.

Creating a detailed report for higher management is not in place which results in higher management not being able to make business improvement decision in a timely fashion.

Manual process also causes challenges for the 3rd party provider like service centers and workshop as they must wait for claim approval amount before they can start work and their payment time is also gets affected as the payments are not being done electronically to the providers as well.

This has resulted in YCompany losing competitive edge in the market as their competitors are providing better claim settlement time and provide constant updates to the customer.

3 Project Requirements

“YCompany” is willing to build customer portal and mobile app which they can use to inform of an accident or loss, they should be able to submit any details regarding accident e.g. photograph of accident or police report and further details. Once the first information is received the claims id should be generated corresponding to the policy and incident. Incident manager, adjustor and surveyor should be notified of the incident. Customer should be constantly updated with the status of the claims processing.

Customer should be provided with a list of partner workshop where they can take the vehicle for repairs. Customer can choose from partner workshop or any external workshop. In case of partner workshop selection customer should be able to set up appointment with the workshop from the customer site itself. The partner workshop should be intimated of the initial accident details. Once the vehicle is dropped off based on the type of policy coverage user should be able to select a rental vehicle from the Car Rental partner from customer site itself. Any change in the status of the vehicle the customer should be notified.

Surveyor should be able to assess the damage on the vehicle once vehicle is in workshop, and should be able to submit the report to back office, Once the assessment is submitted electronically, the adjustor should be able to value the claim to be processed and approve the amount. Once it has been approved electronically Customer should be notified of the amount the claim has been approved for with notification also being sent to the workshop. The workshop should be able to keep updating the status of repairs which should be notified to the customer. The customer should be notified of the any changes in the expected delivery date of the vehicle based on information from workshop. On completion of work the Customer should be able to pay the due amount electronically from the web site itself.

The partner workshop should be able to login to the site and upload work order with respect the car repairs and estimates. Partner Workshop should be able to track their payment based on the incidents they have addressed.

Claims department should be able to generate reports on claims processed, processing time, fraudulent claims, ageing matrix for claim if there are any claims which are pending from long time. Claims department should be able to create claims processed reports based on different roles i.e. Adjustor, Surveyor, Case Manager, Workshop. All of the information interchange should be electronic completely avoiding paper work.

The claims received and processed should be stored in a central document management system for auditing and compliance perspective.

- Customer Portal and App
 - Customer can create a login using their policy details
 - Customer should be able to change the correspondence address
 - Customer should be able to change the billing cycle
 - Customer should be able to make electronic payment.
 - Customer should be able to submit claims.
 - Customer should be able to submit any supporting document
 - Customer should be notified and alerted if any status change happens on claims.
 - Customer should be able to check the Partner Service providers (Repair workshop, authorized service station, Car Rental) based on this location or zip address.
 - Customer should be able to keep track of repair based on work order and progress report submitted by the Repair agency.
- Incident Management
 - Case Management team should be automatically assigned based on the availability of the Surveyor, Adjustor, Case Management also keeping in mind the Surveyor field office covers the area where vehicle is for repairs.
 - Surveyor should be able to submit their assessment online via website or app and once the surveyor assignment is submitted the adjustor should be notified.
 - Adjustor should be able to view the claim, supporting document, assessment report online and should be able to adjudicate the claim based on policy coverage.

- In case Surveyor or Adjustor is not available for claims processing the case manager should be able to delegate the claim processing to other Surveyor or Adjustor.
 - Case Manager should be able to view the complete details of the case and make adjustment to claims which can override the Adjustor and surveyor.
 - Auditor can look at all the details pertaining to the claims and any processing.
- 3rd Party integrations
 - 3rd Party service providers like In network Vehicle Repair Workshop should be able to provided detail work order and estimates with respect to claim.
 - They should be able to provide constant updates with respect to the work that is being done and has been completed.
 - Updates provided by them should be notified to customer (email/SMS) along with the delivery date.
 - They should be able to provide final bill amount for the repairs, which customer should be notified prior of and should be able to pay off from the site.
 - 3rd Party provider should be able to track the claims payment status.
- Internal Users Reporting
 - Internal user should be able to login to the portal.
 - Only Authorized Case Managers should be able to generate reports with respect to the claims they have received.
 - Regional Manager should be able to see the details with respect to claims received in their region with details like
 - Claims Processing Time
 - Claims amount paid out
 - Number of claims with respect to geography.
 - Top management should be able to generate reports based on how each region is processing with respect to claims, which region more claims are being received, which region has higher claims amount paid out.
- Document Management
 - Document Management is the key aspect as every claims document should be saved and archived for further auditing and processing.
- Alerts and Notification

- Alerts and Notification should be sent out constantly to all the relevant parties notifying them of any progress or status change of claims.
- All communication with the customer should be archived for future reference, auditing and compliance.

4 eClaims Application

The proposed eClaims System will essentially provide the following:

- Web application or Mobile App – this web based application and mobile application will be available to authorized users such as customer case managers, auditors, reporting etc.
 - Customer Portal
 - 3rd Party Provider Portal
 - Internal Portal
- Alert and notification system – this sub-system will be analyzing the time to resolve for any incident.
- Reporting
- Document Management

4.1 Web Application

Web application provided by proposed eClaims system will allow legitimate users to access application and perform their desired tasks. Some desired features/functionalities have been mentioned above

4.2 Alert and Notification System

Customer and Concerned Parties should be notified (via SMS/Email) as soon as claims status changes.

5 Design Requirements

Below are some design requirements for eClaims

- Design for evolution, flexibility and reuse
- Componentize functionality as services
- eClaims environment should be up and running 24 x 7. In case of any crashes system should again restart on its own.
- System should be reliable.
- System should have enough logging to help in debug any error condition.

5.1 Functional Requirements

1. The proposed system shall be based on well-defined data and interface standards.
2. Proposed system should utilize an identity management system with role-based user authentication and authorization.
3. Actions the different roles can perform within the system must be configurable without code changes.
4. The proposed system should be able to store, backup, and recover system data in a distributed environment.
5. The system should enforce encryption strategies for sensitive data at rest. Security of confidential business data should be in compliance with the industry standards.
6. The System should be able to take care of non repudiation and also should consider fraudulent claims.

5.2 Non-Functional Requirements

1. System should have flexible design and it should be able to handle increased load in future.
2. It should support both on premise and cloud-based deployment with auto-scaling to match demand.
3. The offered solution must complete 99% of provided services in less than 5000 milliseconds, over both the peak and non-peak hours
4. Proposed solution will have the ability to monitor and maintain the system, and include testability, configurability and upgradeability.
5. Proposed solution must provide guard against various security threats (OWASP Top 10 etc.). System components should utilize industry-proven security standards and protocols.
6. Systems is only in English with no multi-lingual requirements.

6 Artifacts

Following artifacts should be used to complete the assignment.

1. Estimation
<https://nagarro.sharepoint.com/sites/ascoe/pages/detail.aspx?id=1&type=Templates>
2. Solution Approach Design Document

<https://nagarro.sharepoint.com/sites/ascoe/pages/detail.aspx?id=9&type=Templates>

3. DAR Template



DAR Template.docx

7 Assignment

Following is the overall tasks for the assignment of the project.

Sr.	Question	Technology	Details
1	Create a Solution Approach Document containing high level solution design. Note: Use Solution Approach template for this. Pull out NFR in a single page as you think appropriate. You also need to provide the source files (Visio) of the diagrams you have created.		Your solution approach must have following concepts: 1. Detailed solution architecture covering multi-layer design and approach 2. Non Functional Requirements Coverage 3. Technology Stack 4. Performance and Scalability 5. Assumptions & Scope (In Scope/Out Scope) 6. References / Appendix (if any)
2	Create an estimation for the above mentioned selected technology.		Cover all stages of below phases in excel and give final numbers. 1. Requirement Specifications and System Design 2. Implementation 3. Testing

	Note: Use Estimation Sheet Template. Also provide the tentative schedule & resource plan for the project implementation.		4. UAT 5. Documentation, Go-Live, and Transition to operations 6. Project Management and Coordination
3	Create a DAR to define the technology adoption for workflow or document management system and NFR Note: Your document should have detailed comparison of tools / technologies and your recommendation out of those.		DAR Document containing: 1. Identification and explanation of tools for comparison 2. Explanation of each attribute of comparison in detail 3. Comparison Matrix 4. Recommended Tool 5. Risks 6. Assumptions 7. Pricing 8. References
3	A minimal working POC showcasing the key features of the proposed solution.	MEAN/PHP/.Net/J2EE (or any language of your choice)	The evaluation criteria will be focused on below key concepts rather than solely on coding and full fledge implementation: • Design considerations of the application • Layers, tiers and modules of the application • Appropriate project folder structures for various layers like presentation,

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			business, data access etc.
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8 Guidelines

- Take relevant assumptions during the application architecting/designing wherever either requirements are not clear to you or the information required is not provided in this case study.

Following guidelines must be adhered when you create your deliverables.

1. Your deliverables such as documentations etc. must be **Client Ready** presentable format. It means that your deliverables should not have any grammatical mistakes, formatting issues etc.
2. Please try to stick with the format of the templates provided in the section 5.
3. For diagrams, you can use or take hints from AG – architecture design guidelines which you can find at: <http://coe/portfolio/diagram-preparation-guidelines-best-practices>
4. Your POC solution must be zipped after cleanup of all referenced assemblies if there is any. Your solution should be able to build/compile and run after unzipping.